



Q1. Do PWM-based AC control circuits using MOSFETs require ZCD circuits?

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A1. There are two common types of zero-crossing detection (ZCD) circuits: operational amplifiers and passive circuits comprising resistors, capacitors, etc. ZCD circuits are used for detection of zero voltage reference point in an AC signal or in a sinewave signal. A typical zero-crossing detection of the AC signal is shown in Fig. 1.

Many PWM-based AC control circuits such as inverters use MOSFETs and ZCD circuits. MOSFET devices such as CS1501, AOZ5311NQi, AOZ5237Qi and CS1601H have dedicated pins for ZCD and related functions. ZCD circuit is used to initiate the MOSFET switching cycle for monitoring and controlling actions such as synchronisation of the signals.

The ZCD circuit is used in many applications including controlled voltage rectifiers, resonant power supplies, AC motor speed controllers, AC power controllers, among others.

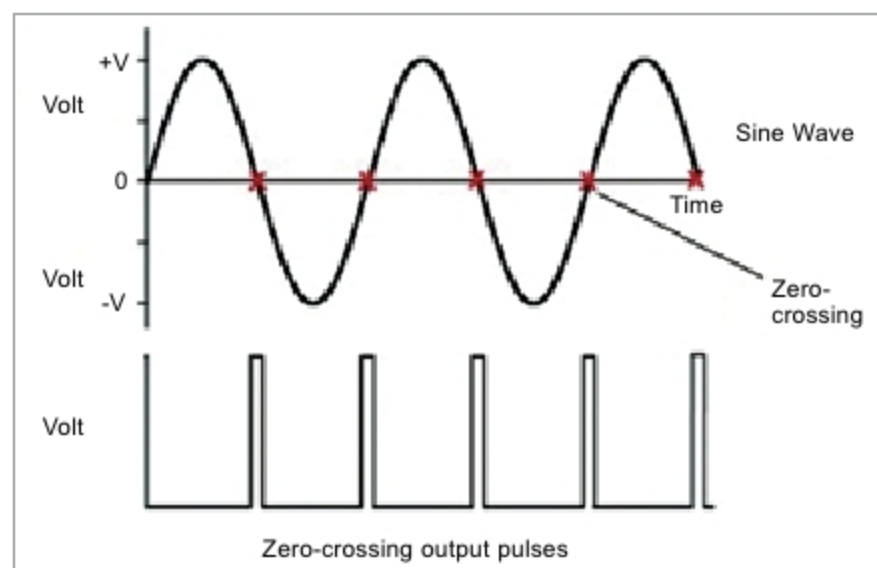


Fig. 1: A typical zero-crossing detection output

Q2. What are real-time clock (RTC) chips? Which are some recent RTC chips launched in the market?

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A2. An RTC is an electronic device, generally an integrated circuit (IC), which keeps track of the current time and maintains accurate time in electronic systems. It is basically like a timepiece that runs on a battery and maintains current time, even when the main power supply goes off.

One of the best examples of its use is in a computer motherboard. Because of this RTC device, we get correct time on our computer/laptop, even after it is shut down and turned on again. RTCs are present in almost all infotainment systems, automotive systems, house metering, consumer electronics, and industrial electronic devices that require clock functionality.

Companies are focusing on miniaturisation of electronic devices for various applications. Therefore small and compact RTC modules are gaining widespread application in a wide range of products. This trend has huge significance for medical monitors and portable terminals.

For the development of ultra-low-power devices, many research efforts are going on to thoroughly understand the dynamics of size and connectivity requirements in RTC devices. Such devices have revolutionised the Internet of Things (IoT) and wearables sector. For instance, recently Seiko Epson Corporation launched two new small RTC modules (RX8111CE and RX4111CE) having low current consumption. RV-8803-C7 RTC chip developed by Micro Crystal is a high-accuracy, ultra-low power device with embedded 32.768kHz crystal.

Next-generation RTC chips with power management and

power switching functions are also available in the market. CBC921xx series chips from Cymbet Corporation are an example of such RTC devices.

Standalone RTC ICs are designed to have backup capabilities by analysing different parameters such as minimum and maximum current consumption in the power supplies. For instance, M41T62 from STMicroelectronics is a standalone RTC with backup that keeps the time with a supply voltage as low as 1.0V and can tolerate voltages of up to 4.5V.

With technological advancements, manufacturers are continuously introducing new features such as embedded memory, timestamp, and watchdog timers. DS3231M and DS3232M RTC chips by Maxim Integrated Products come with such features. DS3231M has temperature-compensated RTC with an internal MEMS resonator and DS3232M has additional memory in the small package. DS3231M RTC chip is shown in Fig. 2.

Key players operating in the global RTC chip market include Maxim Integrated Products, Inc, Seiko Epson Corp, STMicroelectronics, Microchip Technology Inc, NXP Semiconductors, Diodes Incorporated, Texas Instruments Incorporated, Renesas Electronics, Ricoh Semiconductor, ROHM, Micro Crystal, Cymbet and Cypress Semiconductor Corporation.



Fig. 2: An RTC chip (Credit: mouser.com)